

For users looking to create interactive visualisations, Plotly offers a powerful solution. It supports dynamic charts that respond to user interactions, such as zooming, filtering, or selecting specific data points. This interactivity makes Plotly an excellent choice for creating dashboards and web-based reports. Whether you're plotting sales performance over time or visualising real-time sensor data, Plotly provides a user-friendly framework for developing engaging visuals.

For those looking for a no-code solution, Tableau Public is a free tool that allows users to create interactive dashboards and visualisations. Its drag-and-drop interface makes it easy to explore data from different sources and design dashboards without any coding. Tableau is especially popular in business intelligence and marketing, where creating clear and engaging dashboards is important for sharing insights with others. While Tableau Public has some limitations compared to the paid version, it is still a great tool for learning the basics of data visualisation and how to tell a story with data.

For R users, tools like ggplot2 offer great flexibility when it comes to creating high-quality visualisations. ggplot2 is based on the grammar of graphics — a concept that helps users build plots step by step by adding different components, such as data, axes, labels, and colours. This allows users to create very detailed and personalised visualisations. Beginners can start by making simple graphs, like bar charts or scatter plots, while more advanced users can use ggplot2 to create complex, multi-panel plots that are often used in academic research or professional presentations.

Open source: Shaping the future of data science

The future of data science is tightly connected to the growth of open source tools. One of the main reasons open source tools are thriving is community collaboration. Thousands

Must-know data science packages for beginners

Category	Python	R
Automation	AutoML	h2o
Clustering	Scikit-learn	cluster
Data cleaning	Pandas	janitor
Data collection	requests	rvest
Data exploration	Pandas, NumPy	dplyr
Data visualisation	Matplotlib	ggplot2
Deep learning	Keras	keras
Machine learning	Scikit-learn	caret
Model evaluation	Scikit-learn	caret
Statistical analysis	SciPy	stats
Text analytics	TextBlob	text
Text mining and NLP	spaCy	tm
Time series analysis	statsmodels	forecast
Web scraping	Beautiful-Soup	rvest

of active users contribute to the development and improvement of these tools. Platforms like GitHub and Stack Overflow make it easier for data scientists to share ideas and solve problems. For example, scikit-learn and TensorFlow have become successful because their communities continually improve them, keeping them reliable and effective.

Cloud platforms like AWS, Google Cloud, and Microsoft Azure have also transformed the way data scientists work. Cloud solutions allow users to scale projects easily, process massive datasets, and deploy models in real-time. Tools like Apache Spark, Kubernetes, and Jupyter Notebooks are optimised for cloud environments, making collaboration more efficient. As data volumes grow, cloud-integrated open

source tools have become essential for effective analysis. Many cloud platforms even offer free tiers or credits, making it affordable for students and small organisations to access powerful computing resources.

Libraries such as TensorFlow and PyTorch are leading the way in machine learning and AI. These tools are constantly improving, adding new algorithms and enhancing performance. For instance, TensorFlow's Keras API simplifies deep learning, making it beginner-friendly, while PyTorch is popular with researchers due to its flexibility. Tools for natural language processing (NLP) and computer vision, such as spaCy and OpenCV, are also expanding the possibilities for data scientists. These innovations help apply AI across industries like healthcare, finance, and retail.

Automation is becoming an important feature of data science tools. Open source projects like AutoML frameworks, including Auto-sklearn and H2O.ai, automate tasks such as feature selection, hyperparameter tuning, and model evaluation. This allows data scientists to focus more on analysing results and solving problems. Automation also makes it easier for beginners to achieve valuable results without requiring deep technical expertise. As these tools continue to evolve, they will lower the entry barrier for newcomers and promote broader adoption of data science.

Whether you're just starting your tech journey or leading innovation in an organisation, open source tools provide the right help, making it possible for anyone, anywhere, to participate in the exciting field of data science. **END** 🐧

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